Top 108 CAT Data Sufficiency Questions With Video Solutions

Questions

Instructions

Directions for the following four questions: Each question is followed by two statements A and B. Indicate your responses based on data sufficiency

Question 1

The average weight of a class of 100 students is 45 kg. The class consists of two sections, I and II, each with 50 students. The average weight, W_I , of Section I is smaller than the average weight, W_{II} , of Section II. If the heaviest student, say Deepak, of Section II is moved to Section I, and the lightest student, say Poonam, of Section I is moved to Section II, then the average weights of the two sections are switched, i.e., the average weight of Section I becomes W_{II} and that of Section II becomes W_I . What is the weight of Poonam?

A: $W_{II} - W_I = 1.0$

B: Moving Deepak from Section II to I (without any move from I to II) makes the average weights of the two sections equal.

A The question can be answered using A alone but not using B alone.

B The question can be answered using B alone but not using A alone.

C The question can be answered using A and B together, but not using either A or B alone.

D The question cannot be answered even using A and B together.

E None of these

VIEW SOLUTION

Question 2

ABC Corporation is required to maintain at least 400 Kilolitres of water at all times in its factory, in order to meet safety and regulatory requirements. ABC is considering the suitability of a spherical tank with uniform wall thickness for the purpose. The outer diameter of the tank is 10 meters. Is the tank capacity adequate to meet ABC's requirements?

A: The inner diameter of the tank is at least 8 meters.

B: The tank weighs 30,000 kg when empty, and is made of a material with density of 3 gm/cc.

A The question can be answered using A alone but not using B alone.

B The question can be answered using B alone but not using A alone.

C The question can be answered using A and B together, but not using either A or B alone.

D The question cannot be answered even using A and B together.

E None of these

VIEW SOLUTION

Question 3

Consider integers x, y and z. What is the minimum possible value of $x^2 + y^2 + z^2$?

A: x + y + z = 89

B: Among x, y, z two are equal.

A The question can be answered using A alone but not using B alone.

B The question can be answered using B alone but not using A alone.

C The question can be answered using A and B together, but not using either A or B alone.

D The question cannot be answered even using A and B together.

E None of these

VIEW SOLUTION

Question 4

Rahim plans to draw a square JKLM with a point O on the side JK but is not successful. Why is Rahim unable to draw the square?

A: The length of OM is twice that of OL.

B: The length of OM is 4 cm

A The question can be answered using A alone but not using B alone.

B The question can be answered using B alone but not using A alone.

C The question can be answered using A and B together, but not using either A or B alone.

D The question cannot be answered even using A and B together.

E None of these

VIEW SOLUTION

Instructions

Directions for the following four questions:

Each question is followed by two statements, A and B. Answer each question using the following instructions:

Question 5

In a particular school, sixty students were athletes. Ten among them were also among the top academic performers. How many top academic performers were in the school?

A. Sixty per cent of the top academic performers were not athletes.

B. All the top academic performers were not necessarily athletes.

A The question can be answered by using the statement A alone but not by using the statement B alone.

B The question can be answered by using the statement B alone but not by using the statement A alone.

C The question can be answered by using either of the statements alone

D The question can be answered by using both the statements together but not by either of the statements alone.

E The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 6

Five students Atul, Bala, Chetan, Dev and Ernesto were the only ones who participated in a quiz contest. They were ranked based on their scores in the contest. Dev got a higher rank as compared to Ernesto, while Bala got a higher rank as compared to Chetan. Chetan's rank was lower than the median. Who among the five got the highest rank?

A. Atul was the last rank holder.

B. Bala was not among the top two rank holders.

A The question can be answered by using the statement A alone but not by using the statement B alone.

B The question can be answered by using the statement B alone but not by using the statement A alone.

C The question can be answered by using either of the statements alone.

D The question can be answered by using both the statements together but not by either of the statements alone.

E The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 7

Thirty per cent of the employees of a call centre are males. Ten per cent of the female employees have an engineering background. What is the percentage of male employees with engineering background?

A. Twenty five per cent of the employees have engineering background.

B. Number of male employees having an engineering background is 20% more than the number of female employees having an engineering background

A The question can be answered by using the statement A alone but not by using the statement B alone.

B The question can be answered by using the statement B alone but not by using the statement A alone.

C The question can be answered by using either of the statements alone.

D The question can be answered by using both the statements together but not by either of the statements alone.

E The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 8

In a football match, at the half-time, Mahindra and Mahindra Club was trailing by three goals. Did it win the match?

A. In the second-half Mahindra and Mahindra Club scored four goals.

B. The opponent scored four goals in the match.

- **A** The question can be answered by using the statement A alone but not by using the statement B alone.
- **B** The question can be answered by using the statement B alone but not by using the statement A alone.
- **C** The question can be answered by using either of the statements alone.
- **D** The question can be answered by using both the statements together but not by either of the statements alone.
- **E** The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Instructions

In a single elimination tournament, any a player is eliminated with a single loss. The tournament is played in multiple rounds subject to the following rules :

(a) If the number of players, say n, in any round is even, then the players are grouped into n/2 pairs. The players in each pair play a match against each other and the winner moves on to the next round.

(b) If the number of players, say n, in any round is odd, then one of them is given a bye, that is he automatically moves on to the next round. The remaining (n–1) players are grouped into (n–1)/2 pairs. The players in each pair play a match against each other and the winner moves on to the next round. No player gets more than one bye in the entire tournament.

Thus, if n is even, then n/2 players move on to the next round while if n is odd, then (n+1)/2 players move on to the next round. The process is continued till the final round, which obviously is played between two players. The winner in the final round is the champion of the tournament.

Question 9

What is the number of Matches played by the champion?

A. The entry list for the tournament consists of 83 players?

B. The champion received one bye.

- **A** Q can be answered from A alone but not from B alone.
- **B** Q can be answered from B alone but not from A alone.
- **C** Q can be answered from A alone as well as from B alone.
- **D** Q can be answered from A and B together but not from any of them alone.
- **E** Q cannot be answered even from A and B together.

VIEW SOLUTION

Question 10

If the number of players, say n, in the first round was between 65 and 128, then what is the exact value of n?

A. Exactly one player received a bye in the entire tournament.

B. One player received a bye while moving on to the fourth round from the third round.

A Q can be answered from A alone but not from B alone.

B Q can be answered from B alone but not from A alone.

C Q can be answered from A alone as well as from B alone.

D Q can be answered from A and B together but not from any of them alone.

E Q cannot be answered even from A and B together.

VIEW SOLUTION

Instructions

For the following questions answer them individually

Question 11

F and M are father and mother of S, respectively. S has four uncles and three aunts. F has two siblings. The siblings of F and M are unmarried. How many brothers does M have?

A. F has two brothers.

B. M has five siblings.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 12

A game consists of tossing a coin successively. There is an entry fee of Rs. 10 and an additional fee of Re. 1 for each toss of coin. The game is considered to have ended normally when the coin turns heads on two consecutive throws. In this case the player is paid Rs. 100. Alternatively, the player can choose to terminate the game prematurely after any of the tosses. Ram has incurred a loss of Rs. 50 by playing this game. How many times did he toss the coin?

A. The game ended normally.

B. The total number of tails obtained in the game was 138.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 13

Each packet of SOAP costs Rs. 10. Inside each packet is a gift coupon labelled with one of the letters S, O, A and P. If a customer submits four such coupons that make up the word SOAP, the customer gets a free SOAP packets. Ms. X kept buying packet after packet of SOAP till she could get one set of coupons that formed the word SOAP. How many coupons with label P did she get in the above process?

A. The last label obtained by her was S and the total amount spent was Rs. 210.

B. The total number of vowels obtained was 18.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 14

If A and B run a race, then A wins by 60 seconds. If B and C run the same race, then B wins by 30 seconds. Assuming that C maintains a uniform speed what is the time taken by C to finish the race?

A. A and C run the same race and A wins by 375 metres.

B. The length of the race is 1 km.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 15

Is $a^{44} < b^{11}$, given that a = 2 and b is an integer?

A. b is even

B. b is greater than 16

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 16

What are the unique values of b and c in the equation $4x^2 + bx + c = 0$ if one of the roots of the equation is (-1/2)?

A. The second root is 1/2.

B. The ratio of c and b is 1.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 17

AB is a chord of a circle. AB = 5 cm. A tangent parallel to AB touches the minor arc AB at E. What is the radius of the circle?

A. AB is not a diameter of the circle.

B. The distance between AB and the tangent at E is 5 cm.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 18

Is $(1/a^2 + 1/a^4 + 1/a^6 + ...) > (1/a + 1/a^3 + 1/a^5 + ...)$?

A. $0 < a \leq 1$

B. One of the roots of the equation $4x^2 - 4x + 1 = 0$ is a

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 19

D, E, F are the mid points of the sides AB, BC and CA of triangle ABC respectively. What is the area of DEF in square centimeters?

A. AD = 1 cm, DF = 1 cm and perimeter of DEF = 3 cm

B. Perimeter of ABC = 6 cm, AB = 2 cm, and AC = 2 cm.

A The questions can be answered by one of the statements, but not by the other

B The question can be answered using either of the two statements alone.

C The questions can be answered using both the statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both the statements together.

VIEW SOLUTION

Question 20

Zakib spends 30% of his income on his children's education, 20% on recreation and 10% on healthcare. The corresponding percentage for Supriyo are 40%, 25%, and 13%. Who spends more on children's education?

A. Zakib spends more on recreation than Supriyo.

B. Supriyo spends more on healthcare than Zakib.

A The question can be answered by using one of the statements alone but not by using the other statement alone.

B The question can be answered by using either of the statements alone.

C The question can be answered by using both statements together but not by either statement alone.

D The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 21

Four candidates for an award obtain distinct scores in a test. Each of the four casts a vote to choose the winner of the award. The candidate who gets the largest number of votes wins the award. In case of a tie in the voting process, the candidate with the highest score wins the award. Who wins the award?

A. The candidates with top three scores each vote for the top score amongst the other three.

B. The candidate with the lowest score votes for the player with the second highest score.

A. The question can be answered by using one of the statements alone but not by using the other statement alone.

B. The question can be answered by using either of the statements alone.

C. The question can be answered by using both statements together but not by either statement alone.

D. The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 22

In a class of 30 students, Rashmi secured the third rank among the girls, while her brother Kumar studying in the same class secured the sixth rank in the whole class. Between the two, who had a better overall rank?

A. Kumar was among the top 25% of the boys merit list in the class in which 60% were boys.

B. There were three boys among the top five rank holders, and three girls among the top ten rank holders.

A. The question can be answered by using one of the statements alone but not by using the other statement alone.

B. The question can be answered by using either of the statements alone.

C. The question can be answered by using both statements together but not by either statement alone.

D. The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 23

Tarak is standing 2 steps to the left of a red mark and 3 steps to the right of a blue mark. He tosses a coin. If it comes up heads, he moves one step to the right; otherwise he moves one step to the left. He keeps doing this until he reaches one of the two marks, and then he stops.

At which mark does he stop?

A. He stops after 21 coin tosses.

B. He obtains three more tails than heads.

A. The question can be answered by using one of the statements alone but not by using the other statement alone.

B. The question can be answered by using either of the statements alone.

C. The question can be answered by using both statements together but not by either statement alone.

D. The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 24

Ravi spent less than Rs. 75 to buy one kilogram each of potato, onion, and gourd. Which one of the three vegetables bought was the costliest?

A. 2 kg potato and 1 kg gourd cost less than 1 kg potato and 2 kg gourd.

B. 1 kg potato and 2 kg onion together cost the same as 1 kg onion and 2 kg gourd.

A The question can be answered by using one of the statements alone but not by using the other statement alone.

B The question can be answered by using either of the statements alone.

C The question can be answered by using both statements together but not by either statement alone.

D The question cannot be answered on the basis of the two statements.

VIEW SOLUTION

Question 25

Nandini paid for an article using currency notes of denominations Re. 1, Rs. 2, Rs. 5, and Rs. 10 using at least one note of each denomination. The total number of five and ten rupee notes used was one more than the total number of one and two rupee notes used. What was the price of the article?

A. Nandini used a total of 13 currency notes.

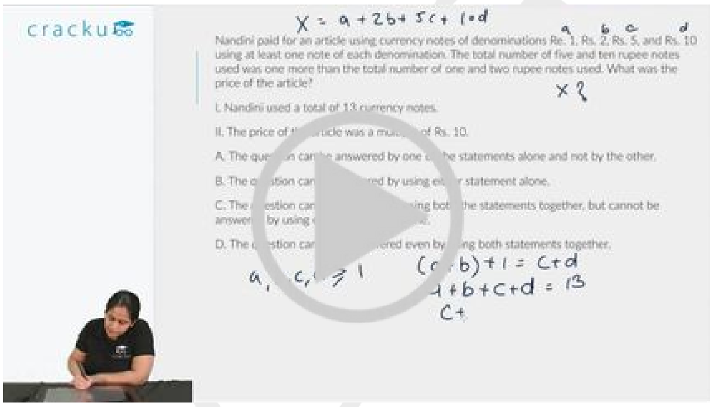
B. The price of the article was a multiple of Rs. 10.

A The question can be answered by using one of the statements alone but not by using the other statement alone.

B The question can be answered by using either of the statements alone.

C The question can be answered by using both statements together but not by either statement alone.

D The question cannot be answered on the basis of the two statements.



VIDEO SOLUTION

Question 26

What are the values of m and n?

I. n is an even integer, m is an odd integer, and m is greater than n.

II. Product of m and n is 30.

A The question can be answered by one of the statements alone and not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 27

Is Country X’s GDP higher than country Y’s GDP?

I. GDPs of the countries X and Y have grown over the past 5 years at compounded annual rate of 5% and 6% respectively.

II. Five years ago, GDP of country X was higher than that of country Y.

A The question can be answered by one of the statements alone and not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 28

What is the value of X?

I. X and Y are unequal positive even integers, less than 10, and $\frac{X}{Y}$ is an odd integer.

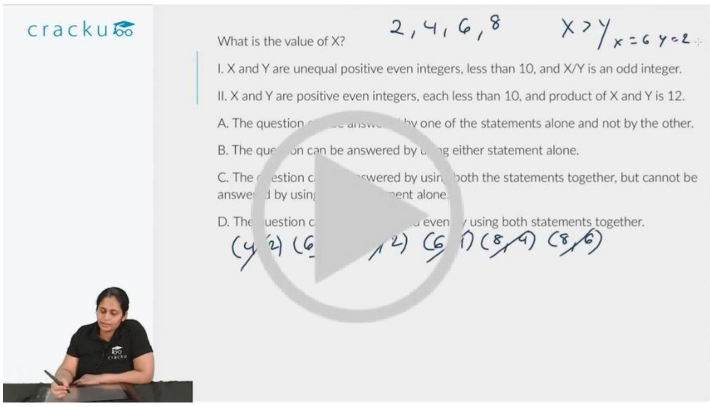
II. X and Y are positive even integers, each less than 10, and product of X and Y is 12.

A The question can be answered by one of the statements alone and not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered even by using both statements together.



VIDEO SOLUTION

Question 29

On a given day a boat ferried 1,500 passengers across the river in 12 hr. How many round trips did it make?

I. The boat can carry 200 passengers at any time.

II. It takes 40 min each way and 20 min of waiting time at each terminal.

- **A** The question can be answered by one of the statements alone and not by the other.
- **B** The question can be answered by using either statement alone.
- **C** The question can be answered by using both the statements together, but cannot be answered by using either statement alone.
- **D** The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 30

What will be the time for downloading software?

I. Transfer rate is 6 kilobytes per second.

II. The size of the software is 4.5 megabytes.

- **A** The question can be answered by one of the statements alone and not by the other.
- **B** The question can be answered by using either statement alone.
- **C** The question can be answered by using both the statements together, but cannot be answered by using either statement alone.
- **D** The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 31

A square is inscribed in a circle. What is the difference between the area of the circle and that of the square?

I. The diameter of the circle is $25\sqrt{2}$ cm.

II. The side of the square is 25 cm.

A The question can be answered by one of the statements alone and not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 32

Two friends, Ram and Gopal, bought apples from a wholesale dealer. How many apples did they buy?

I. Ram bought one-half the number of apples that Gopal bought.

II. The wholesale dealer had a stock of 500 apples.

A The question can be answered by one of the statements alone and not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 33

Consider three real numbers, X, Y, and Z. Is Z the smallest of these numbers?

A. X is greater than at least one of Y and Z.
B. Y is greater than at least one of X and Z.

A if the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B if the question can be answered by using either statement alone.

C if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D if the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 34

Let X be a real number. Is the modulus of X necessarily less than 3?

A. X(X+3)<0

B. X(X-3)>0

A if the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B if the question can be answered by using either statement alone.

C if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D if the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 35

How many people are watching TV programme P?

A. Number of people watching TV programme Q is 1000 and number of people watching both the programmes, P and Q, is 100.

B. Number of people watching either P or Q or both is 1500.

A the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 36

Triangle PQR has angle PRQ equal to 90 degrees. What is the value of PR + RQ?

A. Diameter of the inscribed circle of the triangle PQR is equal to 10 cm.
B. Diameter of the circumscribed circle of the triangle PQR is equal to 18 cm

A the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 37

Harshad bought shares of a company on a certain day, and sold them the next day. While buying and selling he had to pay to the broker one percent of the transaction value of the shares as brokerage. What was the profit earned by him per rupee spent on buying the shares?

A.The sales price per share was 1.05 times that of its purchase price.
B. The number of shares purchased was 100.

A the question can be answered by one of. the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 38

For any two real numbers:

a + b = 1 if both a and b are positive or both a and b are negative.
a + b = -1 if one of the two numbers a and b is positive and the other negative.

What is (2 + 0) + (-5 + -6)?
A. a + b is zero if a is zero
B. a + b = b + a

A the question can be answered by one of. the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 39

There are two straight lines in the x-y plane with equations:

ax + by = c

dx + ey = f

Do the two straight lines intersect?

A. a, b, c, d, e and f are distinct real numbers.

B. c and f are non-zero.

A the question can be answered by one of. the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 40

O is the centre of two concentric circles. AE is a chord of the outer circle and it intersects the inner circle at points B and D. C is a point on the chord in between B and D. What is the value of AC/CE?

A. BC/CD=1
B. A third circle intersects the inner circle at B and D and the point C is on the line joining the centres of the third circle and the inner circle.

A the question can be answered by one of. the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 41

Ghosh Babu has decided to take a non-stop flight from Mumbai to No-man’s-land in South America. He is scheduled to leave Mumbai at 5 am, Indian Standard Time on December 10, 2000. What is the local time at No-man's-land when he reaches there?

A. The average speed of the plane is 700 kilometres per hour.

B. The flight distance is 10,500 kilometres.

A the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 42

What are the ages of two individuals, X and Y?

A. The age difference between them is 6 years.
B. The product of their ages is divisible by 6.

A the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

B the question can be answered by using either statement alone.

C the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 43

In a hockey match, the Indian team was behind by 2 goals with 5 min remaining. Did they win the match?

A. Deepak Thakur, the Indian striker, scored 3 goals in the last 5 min of the match.

B. Korea scored a total of 3 goals in the match.

A The question can be answered by one of the statement alone but not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered by either of the statements.

VIEW SOLUTION

Question 44

Four students were added to a dance class. Would the teacher be able to divide her students evenly into a dance team (or teams) of 8?

A. If 12 students were added, the teacher could put everyone in teams of 8 without any leftovers.

B. The number of students in the class earlier was not divisible by 8.

A The question can be answered by one of the statement alone but not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered by either of the statements.

VIEW SOLUTION

Question 45

Is $x = y$?

A. $(x + y)(1/x + 1/y) = 4$

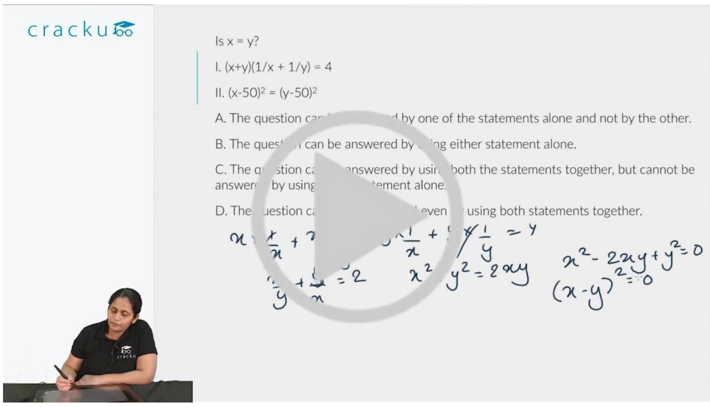
B. $(x - 50)^2 = (y - 50)^2$

A The question can be answered by one of the statement alone but not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered by either of the statements.



VIDEO SOLUTION

Question 46

A dress was initially listed at a price that would have given the store a profit of 20% of the wholesale cost. What was the wholesale cost of the dress?

A. After reducing the listed price by 10%, the dress sold for a net profit of $10.

B. The dress is sold for $50.

A The question can be answered by one of the statement alone but not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered by either of the statements.

VIEW SOLUTION

Question 47

Is 500 the average (arithmetic mean) score in the GMAT?

A. Half of the people who take the GMAT score above 500 and half of the people score below 500.

B. The highest GMAT score is 800 and the lowest score is 200.

A The question can be answered by one of the statement alone but not by the other.

B The question can be answered by using either statement alone.

C The question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D The question cannot be answered by either of the statements.

VIEW SOLUTION

Question 48

Is |x - 2| < 1?

A. |x| < 1

B. |x - 1| < 2

A if the question can be answered by using either statement alone.

B if the question can be answered by one of the statement alone but not by the other.

C if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D if the question cannot be answered by either of the statements.

VIEW SOLUTION

Question 49

People in a club either speak French or Russian or both. Find the number of people in a club who speak only French.

A. There are 300 people in the club and the number of people who speak both French and Russian is 196.

B. The number of people who speak only Russian is 58.

A if the question can be answered by one of the statement alone but not by the other.

B if the question can be answered by using either statement alone.

C if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D if the question cannot be answered by either of the statements.

VIEW SOLUTION

Question 50

A sum of Rs. 38,500 was divided among Jagdish, Punit and Girish. Who received the minimum amount?

A. Jadgish received 2/9 of what Punit and Girish received together.

B. Punit received 3/11 of what Jadgish and Girish received together.

A if the question can be answered by one of the statement alone but not by the other.

B if the question can be answered by using either statement alone.

C if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

D if the question cannot be answered by either of the statements.

VIEW SOLUTION

Question 51

If R is an integer between 1 & 9, P - R = 2370, what is the value of R?
I. P is divisible by 4.
II. P is divisible by 9.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 52

A man distributed 43 chocolates to his children. How many of his children are more than five years old?
I. A child older than five years gets 5 chocolates.
II. A child 5 years or younger in age gets 6 chocolates.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 53

Ramu went by car from Calcutta to Trivandrum via Madras, without any stoppages. The average speeds for the entire journey was 40 kmph. What was the average speed from Madras to Trivandrum?
I. The distance from Madras to Trivandrum is 0.30 times the distance from Calcutta to Madras.
II. The average speed from Madras to Trivandrum was twice that of the average speed from Calcutta to Madras.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 54

x, y, and z are three positive odd integers, Is x+z divisible by 4?
I. y - x = 2
II. z - y = 2

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 55

The unit price of product P1 is non-increasing and that of product P2 is decreasing. Which product will be costlier 5 years hence?
I. Current unit price of P1 is twice that of P2.
II. 5 years ago, unit price of P2 was twice that of P1.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 56

X is older than Y, Z is younger than W and V is as old as Y. Is Z younger than X?
I. W may not be older than V
II. W is not older than V

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 57

How long did Mr. X take to cover 5000 km. journey with 10 stopovers?
I. The i^{th} stopover lasted i^2 minutes.
II. The average speed between any two stopovers was 66 kmph.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 58

Is $[(x^{-1} - y^{-1})/(x^{-2} - y^{-2}] > 1$?
I. x + y > 0
II. x and y are positive integers and each is greater than 2.

A The question can be answered with the help of statement I alone,

B The question can be answered with the help of statement II alone,

C Both, statement I and statement II are needed to answer the question, and

D The statement cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 59

Is it more profitable for Company M to produce Q?
I. Product R sells at a price four times that of Q
II. One unit of Q requires 2 units of labour, while one unit of R requires 5 units of labour. There is a no other constraint on production.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 60

A train started from Station A, developed engine trouble and reached Station B, 40 minutes late. What is the distance between Stations A and B?
I. The engine trouble developed after travelling 40 km from Station A and the speed reduced to 1/4th of the

original speed.
II. The engine trouble developed after travelling 40 km from station A in two hours and the speed reduced to 1/4th of the original speed.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 61

What is the value of prime number x?
I. $x^2 + x$ is a two-digit number greater than 50
II. x^3 is a three digit number.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 62

The average of three unequal quotations for a particular share is Rs.110. If all are quoted in integral values of rupee, does the highest quotation exceed Rs. 129?
I. The lowest quotation Rs. 100.
II. One of the quotations is Rs. 115.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 63

How many people (from the group surveyed) read both Indian Express and Times of India?
I. Out of total of 200 readers, 100 read Indian Express, 120 read Times of India and 50 read Hindu.
II. Out of a total of 200 readers, 100 read Indian Express, 120 reads Times of India and 50 read neither.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question.

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 64

X says to Y, 'I am 3 times as old as you were 3 years ago'. How old is X?
I. Y's age 17 years from now is same as X.s present age.
II. X's age nine years from now is 3 times Y.s present age.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 65

What is the radius of the circle?
I. Ratio of its area to circumference is > 7.
II. Diameter of the circle is <=32.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 66

What is the time difference between New York and London?
I. The departure time at New York is exactly 9.00 a.m local time and the arrival time at London is at 10.00 a.m. local time.
II. The flight time is 5 hours.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 67

Mr. Murthy takes the morning train to his office from station A to station B, and his colleague Mr.Rahman joins him on the way. There are three stations C, D and E on the way not necessarily in that sequence. What is the sequence of stations?
I. Mr. Rahman boards the train at D.
II. Mr. Thomas, who travels between C & D has two segments of journey in common with Mr. Murthy but none with Mr. Rahman.

A The question can be answered with the help of statement I alone.

B The question can be answered with the help of statement II, alone.

C Both, statement I and statement II are needed to answer the question

D The question cannot be answered even with the help of both the statements.

VIEW SOLUTION

Question 68

What is the number of type-2 widgets produced, if the total number of widgets produced is 20,000? The company produces only two types of widgets.
I. If the production of type-1 widgets increases by 10% and that of type-2 decreases by 6%, the total production remains the same.
II. The ratio in which type-1 and type-2 widgets are produced is 2 : 1.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 69

What is the value of $a^3 + b^3$?
I. $a^2 + b^2 = 22$
II. $ab = 3$

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 70

Is the number completely divisible by 99?
I. The number is divisible by 9 and 11 simultaneously.
II. If the digits of the number are reversed, the number is divisible by 9 and 11.

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 71

A person is walking from Mali to Pali, which lies to its north-east. What is the distance between Mali and Pali?

I. When the person has covered $\frac{1}{3}$ the distance, he is 3 km east and 1 km north of Mali.

II. When the person has covered $\frac{2}{3}$ the distance, he is 6 km east and 2 km north of Mali.

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 72

What is the value of x and y?
I. $3x + 2y = 45$
II. $10.5x + 7y = 157.5$

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 73

Three friends P, Q and R are wearing hats, either black or white. Each person can see the hats of the other two persons. What is the colour of P's hat? I. P says that he can see one black hat and one white hat. II. Q says that he can see one white hat and one black hat.

- **A** The question can be answered with the help of one statement alone.
- **B** The question can be answered with the help of any one statement independently.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 74

What is the speed of the car?

I. The speed of a car is 10 (km/hr) more than that of a motorcycle.
II. The motorcycle takes 2 hr more than the car to cover 100 km.

- **A** The question can be answered with the help of one statement alone.
- **B** The question can be answered with the help of any one statement independently.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 75

What is the ratio of the volume of the given right circular cone to the one obtained from it?
I. The smaller cone is obtained by passing a plane parallel to the base and dividing the original height in the ratio 1 : 2.
II. The height and the base of the new cone are one-third those of the original cone.

- **A** The question can be answered with the help of one statement alone.
- **B** The question can be answered with the help of any one statement independently.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 76

What is the area bounded by the two lines and the coordinate axes in the first quadrant?
I. The lines intersect at a point which also lies on the lines 3x - 4y = 1 and 7x - 8y = 5.
II. The lines are perpendicular, and one of them intersects the Y-axis at an intercept of 4.

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 77

What is the cost price of the chair?
I. The chair and the table are sold at profits of 15% and 20% respectively.
II. If the cost price of the chair is increased by 10% and that of the table is increased by 20%, the profit reduces by Rs. 20.

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 78

After what time will the two persons Tez and Gati meet while moving around the circular track? Both of them start at the same point and at the same time.

I. Tez moves at a constant speed of 5 m/s, while Gati starts at a speed of 2 m/s and increases his speed by 0.5 m/s at the end of every second thereafter.

II. Gati can complete one entire lap in exactly 10 s.

A The question can be answered with the help of one statement alone.

B The question can be answered with the help of any one statement independently.

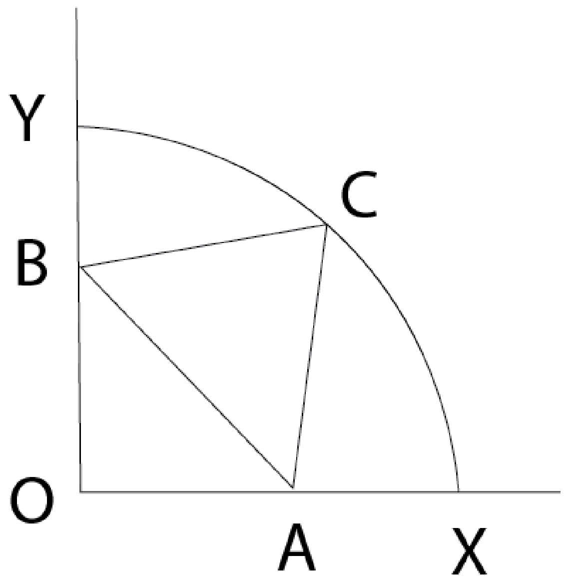
C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 79

Find the length of AB if $\angle YBC = \angle CAX = \angle YOX = 90$.



I. Radius of the arc is given.

II. OA = 5

- **A** The question can be answered with the help of any one statement alone but not by the other statement.
- **B** The question can be answered with the help of either of the statements taken individually.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 80

Is n odd?

I. n is divisible by 3, 5, 7 and 9.

II. $0 < n < 400$

- **A** The question can be answered with the help of any one statement alone but not by the other statement.
- **B** The question can be answered with the help of either of the statements taken individually.
- **C** The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 81

Find $2 \circledast 3$, where $2 \circledast 3$ need not be equal to $3 \circledast 2$

I. $1 \circledast 2$=3

II. $a \circledast b = \frac{a+b}{a}$, where a and b are positive.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 82

Radha and Rani appeared in an examination. What was the total number of questions?

I. Radha and Rani together solved 20% of the paper.

II. Radha alone solved 5/3 rd of the paper solved by Rani.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 83

What is the price of tea?

I. Price of coffee is Rs. 5 more than that of tea.

II. Price of coffee was Rs. 5 less than the price of a cold drink which cost three times the price of tea.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 84

What is the value of ‘a’?

I. Ratio of a and b is 3 : 5, where b is positive.

II. Ratio of 2a and b is 10 12 , where a is positive.

- **A** The question can be answered with the help of any one statement alone but not by the other statement.
- **B** The question can be answered with the help of either of the statements taken individually.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 85

In a group of 150 students, find the number of girls.

I. Each girl was given 50 paise, while each boy was given 25 paise to purchase goods totaling Rs. 49.

II. Girls and boys were given 30 paise each to buy goods totalling Rs. 45.

- **A** The question can be answered with the help of any one statement alone but not by the other statement.
- **B** The question can be answered with the help of either of the statements taken individually.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 86

There are four envelopes — E1, E2, E3 and E4 — in which one was supposed to put letters L1, L2, L3 and L4 meant for persons C1, C2, C3 and C4 respectively, but by mistake the letters got jumbled up and went in wrong envelopes. Now if C2 is allowed to open an envelope at random, then how will he identify the envelope containing the letter for him?

I. L2 has been put in E1.

II. The letter belonging to C3 has gone in the correct envelope.

- **A** The question can be answered with the help of any one statement alone but not by the other statement.
- **B** The question can be answered with the help of either of the statements taken individually.
- **C** The question can be answered with the help of both statements together.
- **D** The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 87

There are four racks numbered 1, 2, 3, 4 and four books numbered 1, 2, 3, 4. If an even rack has to contain an odd-numbered book and an odd rack contains an even-numbered book, then what is the position of book 4?

I. Second book has been put in third rack.

II. Third book has been put in second rack.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 88

Find the value of X in terms of ‘a’.

I. Arithmetic mean of X and Y is ’a’ while the geometric mean is also ‘a’.

II. Y X = R; X - Y = D.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Question 89

There are two concentric circles C1 and C2 with radii r1 and r2. The circles are such that C1 fully encloses C2. Then what is the radius of C1?

I. The difference of their circumference is k cm.

II. The difference of their areas is m sq. cm.

A The question can be answered with the help of any one statement alone but not by the other statement.

B The question can be answered with the help of either of the statements taken individually.

C The question can be answered with the help of both statements together.

D The question cannot be answered even with the help of both statements together.

VIEW SOLUTION

Instructions

Each question is followed by two statements, I and II. Mark the answer

Question 90

A tractor travelled a distance 5 m. What is the radius of the rear wheel?
I. The front wheel rotates ‘N’ times more than the rear wheel over this distance.
II. The circumference of the rear wheel is ‘t’ times that of the front wheel.

- **A** if the question cannot be answered even with the help of both the statements taken together.
- **B** if the question can be answered by any one of the two statements.
- **C** if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.
- **D** if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 91

What is the ratio of the two liquids A and B in the mixture finally, if these two liquids kept in three vessels are mixed together? (The containers are of equal volume.)
I. The ratio of liquid A to liquid B in the first and second vessel is 3 : 5, 2 : 3 respectively.
II. The ratio of liquid A to liquid B in vessel 3 is 4:3.

- **A** if the question cannot be answered even with the help of both the statements taken together.
- **B** if the question can be answered by any one of the two statements.
- **C** if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.
- **D** if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 92

If a,b and c areintegers,is(a-b+c)>(a+b-c)?
I. b is negative.
II. c is positive.

- **A** if the question cannot be answered even with the help of both the statements taken together.
- **B** if the question can be answered by any one of the two statements.
- **C** if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 93

If α and β are the roots of the equation $(ax^2 + bx + c = 0)$, then what is the value of $(\alpha^2 + \beta^2)$?
I.$\alpha + \beta = \frac{-b}{a}$
II.$\alpha\beta = \frac{c}{a}$

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 94

What is the cost price of the article?
I. After selling the article, a loss of 25% on cost price is incurred.
II. The selling price is three-fourths of the cost price.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 95

What is the selling price of the article?
I. The profit on sales is 20%.
II. The profit on each unit is 25% and the cost price is Rs. 250.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 96

How many different triangles can be formed?
I. There are 16 coplanar, straight lines.
II. No two lines are parallel.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 97

What is the total worth of Lakhiram's assets?
I. A compound interest at 10% on his assets, followed by a tax of 4% on the interest, fetches him Rs. 1,500 this year.
II. The interest is compounded once every four months.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Question 98

How old is Sachin in 1997?
I. Sachin is 11 years younger than Anil whose age will be a prime number in 1998.
II. Anil's age was a prime number in 1996.

A if the question cannot be answered even with the help of both the statements taken together.

B if the question can be answered by any one of the two statements.

C if each statement alone is sufficient to answer the question, but not the other one (e.g. statement I alone is required to answer the question, but not statement II and vice versa.

D if both statements I and II together are needed to answer the question.

VIEW SOLUTION

Instructions

Each question is followed by two statements, A and B. Choose the option based on the given statements and questions

Question 99

The average weight of students in a class is 50 kg. What is the number of students in the class?

A. The heaviest and the lightest members of the class weigh 60 kg and 40 kg respectively.

B. Exclusion of the heaviest and the lightest members from the class does not change the average weight of the students.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 100

A small storage tank is spherical in shape. What is the storage volume of the tank?

A. The wall thickness of the tank is 1 cm.

B. When the empty spherical tank is immersed in a large tank filled with water,20 litres of water overflow from the large tank.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 101

Mr. X starts walking northwards along the boundary of a field, from point A on the boundary, and after walking for 150 metres reaches B, and then walks westwards, again along the boundary, for another 100 metres when he reaches C. What is the maximum distance between any pair of points on the boundary of the field?

A.The field is rectangular in shape.

B.The field is a polygon, with C as one of its vertices and A the mid point of a side.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 102

A line graph on a graph sheet shows the revenue for each year from 1990 through 1998 by points and joins the successive points by straight line segments. The point for revenue of 1990 is labelled A, that for 1991 as B, and that for 1992 as C. What is the ratio of growth in revenue between 91-92 and 90-91?

A.The angle between AB and X-axis when measured with a protractor is 40 degrees, and the angle between CB and X-axis is 80 degrees.

B.The scale of Y-axis is 1 cm = 1000 Rs.

A Choose 1 if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B Choose 2 if the question can be answered by using either statement alone.

C Choose 3 if the question can be answered by using both statements together, but cannot be answered using either statement alone.

D Choose 4 if the question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 103

There is a circle with centre C at the origin and radius r cm. Two tangents are drawn from an external point D at a distance d cm from the centre. What are the angles between each tangent and the X-axis?

A.The coordinates of D are given

B.The X-axis bisects one of the tangents.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 104

Find a pair of real numbers x and y that satisfy the following two equations simultaneously. It is known that the values of a, b, c, d, e and f are non-zero. ax + by = c dx + ey = f

A. a = kd and b = ke, c = kf, k is not equal to 0

B. a = b = 1, d = e = 2, f is not equal to 2c

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 105

Three professors A, B and C are separately given three sets of numbers to add. They were expected to find the answers to 1+1, 1+1+2, and 1+1 respectively. Their respective answers were 3, 3, and b How many of the professors are mathematicians?

A. A mathematician can never add two numbers correctly, but can always add three numbers correctly.

B. When a mathematician makes a mistake in a sum, the error is + l or -1.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 106

How many among the four students A, B, C and D have passed the exam'?

A.The following is a true statement: A and B passed the exam.

B.The following is a false statement: At least one among C and D has passed the exam.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 107

What is the distance x between two cities A and B in integral number of Kms?

A.x satisfies the equation $\log_2 x = \sqrt{x}$

B. x less than or equal to 10 Kms

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Question 108

Mr. Mendel grew one hundred flowering plants from black seeds and white seeds, each seed giving rise to one plant. A plant gives flowers of only one colour. From a black seed comes a plant giving red or blue flowers. From a white seed comes a plant giving red or white flowers. How many black seeds were used by Mr. Mendel?

A. The number of plants with white flowers was 10.

B. The number of plants with red flowers was 70.

A The question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

B The question can be answered by using either statement alone.

C The question can be answered by using both statements together, but cannot be answered using either statement alone.

D The question cannot be answered even by using both statements together.

VIEW SOLUTION

Answers

1.**C**	2.**B**	3.**A**	4.**A**	5.**A**	6.**D**	7.**C**	8.**E**

9.**D**	10.**D**	11.**A**	12.**B**	13.**C**	14.**C**	15.**A**	16.**B**
17.**A**	18.**A**	19.**B**	20.**A**	21.**A**	22.**A**	23.**B**	24.**C**
25.**D**	26.**C**	27.**D**	28.**A**	29.**A**	30.**C**	31.**B**	32.**D**
33.**C**	34.**A**	35.**C**	36.**C**	37.**A**	38.**C**	39.**D**	40.**B**
41.**D**	42.**D**	43.**D**	44.**A**	45.**A**	46.**A**	47.**D**	48.**B**
49.**C**	50.**C**	51.**B**	52.**C**	53.**C**	54.**C**	55.**D**	56.**B**
57.**C**	58.**B**	59.**C**	60.**B**	61.**A**	62.**A**	63.**B**	64.**A**
65.**D**	66.**C**	67.**C**	68.**B**	69.**D**	70.**B**	71.**B**	72.**D**
73.**D**	74.**C**	75.**B**	76.**C**	77.**D**	78.**D**	79.**A**	80.**C**
81.**A**	82.**D**	83.**C**	84.**D**	85.**A**	86.**A**	87.**A**	88.**A**
89.**C**	90.**A**	91.**D**	92.**D**	93.**D**	94.**A**	95.**C**	96.**A**
97.**D**	98.**A**	99.**D**	100.**C**	101.**C**	102.**A**	103.**B**	104.**A**
105.**D**	106.**C**	107.**C**	108.**D**				

Explanations

1.**C**
Let w1 and w2 be average of both the groups respectively.

Since average of whole class is 45. $\frac{50*w1+50*w2}{100} = 45$ => w1 + w2 = 90

And if we consider case A we have w2-w1 = 1. From the two equations average weight can be found out.

Further using the equations deduced from given condition which are : $\frac{50*w1-l+h}{50} = w2$ and $\frac{50*w2+l-h}{50} = w1$ where l and h is weight of poonam and deepak respectively. However, both of the above equations are effectively the same. Hence we have two equations and 3 unknowns, thus we cannot find the weights by using statement I alone.

If we consider statement B, then we can get another equation. So we will have 3 equations and then we can solve them for the 3 variables.
So both the equations are required to answer the question.

VIEW SOLUTION

2.**B**
Using statement A alone, we can determine the inner volume of the tank to be at least 4/3 π (125-64) = 256 m^3
.
But we do not know if it is more than 400 m^3. So, using statement A alone, we cannot determine the answer.
Using statement B alone, volume of tank = 30000 kg / 3gm/cc = $10^7 * 10^{-6}$ = 10 m^3
So, inner volume = 4/3 π 125 - 10 > 400 m^3
So, using statement B alone, we can answer the question

VIEW SOLUTION

3.**A**
We know that the $x^2 + y^2 + z^2$ will be minimum when x = y = z = 89/3
Since that is not an integer, we can consider integer values that are closest to 89/3
Let x = y = 30 and z = 29 => $x^2 + y^2 + z^2$ = 2641
If x = y = 29 and z = 31,$x^2 + y^2 + z^2$ = 2643
So, the minimum is 2641
So, the question can be answered using statment 1 alone
Using statement 2 alone, we cannot answer the question.
Option a)

VIEW SOLUTION

4. **A**

If the side of the square is x cm, then the maximum length of OM is $\sqrt{2}x$

The minimum length of OL is x.

So, OM can never be 2 times OL

So, using statement A alone, we can conclude that Rahim is unable to draw the square.

Using statement B alone, we cannot answer the question.

So, option a) is the correct answer.

VIEW SOLUTION

5. **A**

No. of athletes which are top academic performers: 10

According to statement a) Sixty per cent of the top academic performers were not athletes so 40% of top academic performers are athletes which we know is 10. So total no. of top academic performers are 25.

Hence statement a is alone to answer the question.

while we cant deduce anything from statement b .

VIEW SOLUTION

6. **D**

From the information given in the question, D has a better rank than E and B has a better rank than C. Also, C = 4 or 5.

From statement A, A = 5 => C = 4. But we do not know if D or B is number 1.

From statement B, B = 3 or 4. Again, we cannot determine the number 1 rank precisely.

By using both the statements together, we know C = 4, A = 5, B = 3 and D has a better rank than E.

So, D = 1 and E = 2. Therefore, the question can be answered by using both the statements together.

Hence, option D.

VIEW SOLUTION

7. **C**

Let the total number of employees be 100k.

Number of males = 30k. Number of females = 70k.

Number of female engineers = 7k.

From statement A, number of male engineers = 25k - 7k = 18k.

From statement B, number of male engineers = (6/5)*7k.

Therefore, percentage of male engineers can be calculated.

So, the question can be answered by either statement alone.

VIEW SOLUTION

8. **E**

From statement A, we know that the club scored four goals in the second half but we do not know the number of goals scored by the opposition. So, we cannot answer the question.

From statement B, we know that the opposition scored 4 goals in the match but we do not know the number of goals scored by Mahindra and Mahindra club. So, we cannot answer the question.

Using both the statements, we know that the opposition scored 4 goals in the match and Mahindra and

Mahindra scored 4 goals in the second half. But, we do not know if the score was 3-0 or 4-1 at the end of first half. If it was 4-1, then Mahindra and Mahindra would win the match after the second half. But if the score was 3-0 at the end of first half, the match would end in a draw after the second half.
Therefore, the question cannot be answered even by using both the statements together.

VIEW SOLUTION

9. **D**
If we consider statement a then 2 possibilities occur , the champion may or may not receive bye and the answer for both possibilities would be different.
If we consider both statements a & b together we can find the exact answer to the question.

VIEW SOLUTION

10. **D**
Using only statement A, we cannot say anything because the player might have received the bye in any round and the total number of players differs if the round in which the player got a bye changes.

Using only statement B, we cannot saty anything because there might be players who received a bye in other rounds.

Using both the statements, we know that only 1 player got a bye in the whole tournament and that happened in round 3.

If the number of players were between 65 and 128, then the number of rounds is 7.

=> In the fourth round there would be 16 people.

One person got a bye here => 31 people in round 3 => 62 people in round 2 => 124 people in round 1.

Hence, we can ansswer the question using both the statements together.

VIEW SOLUTION

11. **A**
Statement B is redundant. We already know that M has 5 siblings from the question. From statement A alone, we know that F has 2 brothers and M has 2 brothers and 3 sisters.

VIEW SOLUTION

12. **B**
Considering statement A. The game ended normally. Here we know that last 2 results were heads and the person receives 100 Rs. and since there was overall loss of 50 Rs we can calculate no. of matches which ended in tails or non consecutive heads . hence , statement a is sufficient to answer.
Considering statement B. The total number of tails obtained in the game was 138. So we know person lost here rs.138 and we can add any no. of single heads in between those tails such that the overall loss will be 50Rs. Hence this statement is also sufficient to answer.

VIEW SOLUTION

13. **C**
Considering statements a we can deduce that she got letter S in her 21st purchase in previous 20 puchase she had either A, O or P . Hence statement a is not enough to answer. If we consider statement b we will know that out of 20 purchases 18 are vowel so 2 purchases contained letter P . So both statements are necessary to

answer the question.

VIEW SOLUTION

14. **C**

Let the distance be x. Time taken by A is T, so time taken by B is T+60 and time taken by C is T+90.
From statement A, we know that the ratio of speeds of A and C is x:(x-375) = x/T : x/(T+90)
We have only 1 equation and 2 unknowns. We need the second statement to find the values. So, the question can be answered by using both the statements together.

VIEW SOLUTION

15. **A**

If b = 4, then the answer to the question is no. If b = 32, then the answer is yes. So, using statement A alone, we cannot answer the question.
Using statement B alone, we can conclude that $a^{44} < b^{11}$

VIEW SOLUTION

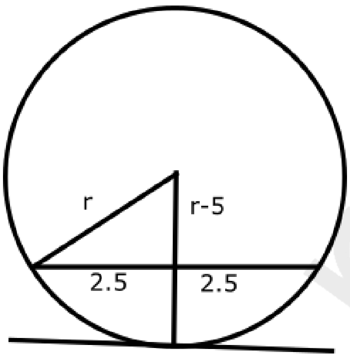
16. **B**

Using statement A, sum of roots = 0 and product of roots = 0, so b = c = 0.
Using statement B, sum of roots = x - 1/2 = -b/4 and product of roots = c/4 = b/4 = -x/2
So, we can calculate the values of b and c using either statement alone

VIEW SOLUTION

17. **A**

Let the radius be r. Using statement B alone, r, r-5 and 2.5 form a right-angled triangle. So, we can answer the question using statement B alone.



VIEW SOLUTION

18. **A**

Consider the first statement:
When the common ratio is less than 1 we can apply the formula of sum of infinite terms.
So, LHS = $1/(a^2 - 1)$
RHS = $a/(a^2 - 1)$
If a<1 then LHS<RHS

If a = 1,then LHS = RHS

So, we cannot answer the question using statement 1 alone

Using statement 2 alone, we know that a = 1/2. So, RHS > LHS
Hence, option a)

VIEW SOLUTION

19. **B**
From statement 1 alone, we can infer that the triangle ABC is an equilateral triangle with side = 2 cm
Similarly, from statement 2 alone, we can infer that the triangle ABC is an equilateral of side 2 cm
So, the question can be answered using either statement alone

VIEW SOLUTION

20. **A**
We have to find out if 3Z > 4S => Z > 1.33S

According to statement 1, Z/5 > S/4 => Z > 1.25S

According to statement 2, 13S > 10Z => Z < 1.3S

So, using statement 2 alone, we can answer the question

VIEW SOLUTION

21. **A**
Let the persons be A,B,C and D in order of the scores tha is, A scores the highest and D the least.

Statement A: A votes for B , B and C votes for A
Even if D votes for B,then also A wins as A has higher score as compared to B.
A alone is sufficient.

StatementB: D votes for B
Nothing can be inferred from this statement.

VIEW SOLUTION

22. **A**
We know that Rashmi secured the third rank among the girls, while her brother Kumar studying in the same class secured the sixth rank in the whole class.

Considering statement A we have 18 B and 12 G and out of 18 boys kumar is among top 4 . But in this case we can't get the relative posiiton of rashmi and kumar.

Consider statement B. We know that among top 5 , 3 are boys and 2 are girls .So rashmi is not among top5 and we know that Kumar's rank is 6 . So Rashmi is definately below Kumar. Hence, the question can be answered by using one of the statements alone but not by using the other statement alone.

VIEW SOLUTION

23. **B**
Consider Statement A:

B_ _x_R

If the person has to reach red, then he has to take 2 steps in the most simplest manner. Now suppose he first takes left step and then tkes a right step. He has taken 2 steps and has to take further 2 steps in order to reach red. In this way, a person has to take even number of steps to reach red. In the similar manner, a person has to take three steps to reach blue in the simplest manner. It means he would have to take odd number of steps to reach blue and since 21 is an odd number, therefore the person would reach blue step.

Consider Statement B,

It is given that the person obtains 3 more tails as compared to heads. Take the case of 1 head and four tails. In this case, he would reach blue step.Now take the case of 2 heads and 5 tails. In this case 2 heads are neutralised by 2 tails. The ramaining portion is 3 tails which would help to reach blue steps. In this manner, no matter how many heads are, three more tails would make sure that the person reaches blue corner.

VIEW SOLUTION

24. **C**
Using statement 1 alone, 2P + G < P + 2G

=> P < G

Using statement 2 alone, P + 2O = O + 2G

=> P + O = 2G

So, using either statement alone, we cannot find the answer to the question.

However, by using both the statements together, we can determine that P < G < O

So, option c) is the answer.

VIEW SOLUTION

25. **D**
Let the no. of 1,2,5,10 Rs notes be x,y,z,w.

According to given condition we have z+w=x+y+1 .

Now considering statement A

The possible values are 4+3=3+3+1 .

But in this case the data is not sufficient to find price of article as the both 5 and 10 rs can have any of the 4 or 3 notes.

Hence statement A is not sufficient.

Now considering statement B ,

The price of the article was a multiple of Rs. 10.

For this, more than 1 possibility of no. of notes of a particular value is there. Also 2 statements together can't give the total price of the article.

VIDEO SOLUTION

26. **C**
Using either statement alone, we cannot answer the question.
The factors of 30 are 1, 2, 3, 5, 6, 10, 15 and 30. From the information given in both statements, the value of m is 15, and n is 2. So, option c).

This is a previous year's CAT question. We can also consider the negative values, which gives us other possibilities. So, a single answer can't be obtained.

VIEW SOLUTION

27. **D**
Even by using both the statements, we do not know whether country X has a higher GDP than country Y because we do not know the absolute values of the GDPs 5 years back. So, the question cannot be answered.

VIEW SOLUTION

28. **A**
If we take the first statement alone,
X and Y can be among 2, 4, 6, and 8.
It is given that X/Y is an odd integer. So, X must be greater than Y.
If we take different pairs of (X,Y) among (8, 2), (8, 4), (8, 6), (6, 2), (6, 4), (4, 2), only (6, 2) satisfies the above condition of X/Y being an odd integer.
So, we can uniquely determine the values of X and Y which are 6 and 2 respectively.

If we take the second statement alone,
12 = 2 * 2 * 3
We want 12 to be the product of two even numbers.
The only possibility is when the numbers are 2 and 6.
However, we don't know any relation between X and Y.
So, both of them can be either 2 or 6.
Therefore, statement II alone is not sufficient to determine the value of X.

Hence, option A is the correct answer.

VIDEO SOLUTION

29. **A**
The question asked is about the number of round trips made in 12 hours. This can be answered if we know the time of travel and the time of waiting. So, the question can be answered by using statement 2 alone but not by using statement 1 alone.

VIEW SOLUTION

30. **C**
We cannot answer the question using either of the statements alone. If we consider both the statements i.e. transfer rate is 6 kilobytes per second and the size of the software is 4.5*1024 kilobytes = 4608 kilobytes then download time = 4608 / 6 seconds. Hence, the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

VIEW SOLUTION

31. **B**
In these type of questions, we don't need to find out the exact answer but just check if the answer can be calculated.

When a square is inscribed inside a circle, the diagonal of the square is a diameter of the circle. Hence, if either the circle diameter or side of the square is given, then we can determine the areas of both the circle and square as well as the difference between their areas.

Hence, the question can be answered by using either statement alone.

VIEW SOLUTION

32. **D**
Consider the first statement. We just know the relation between number of apples of bought by the two guys. Hence, the question cannot be answered using this statement.

Based on the total number of available apples, we cannot determine how many of them were bought by Ram and Gopal.

Using both statements together, we don't know how many apples the two of them bought together. Hence, we cannot answer the question using even both the statements together.

VIEW SOLUTION

33. **C**
If we consider statemnet A. X is greater than at least one of Y and Z. Here X>Z or X>Y, both are possible.

By statement b. Y is greater than at least one of X and Z. Here Y>X or Y>Z.

If we consider bith the statements together then X>Y>Z or X,Y>Z.

In both cases Z is the smallest.

VIEW SOLUTION

34. **A**
Considering statement A, X(X+3)<0 . Here values of x which satisfy are X=-2 , -1 .
Considering statement B, X(X-3)>0 . Here values of x which satisfy are X = ..,-2,-1,4,5,6,... So using only statement 1 we know that uniquely modulus of x has to be less than 3. Hence, the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone.

VIEW SOLUTION

35. **C**
Considering 1st statement we know that number of people watching Q alone is 900 and both P and Q is 100. But we cant find the number of people watching P. Also, using statement B alone, we cannot find the number of people watching P. But if we consider B along with A , we can definately get total no. of people watching P . Hence, the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

VIEW SOLUTION

36. **C**
Let a,b and c be length of PR, QR and PQ.

By 2nd statement we know that c=18 and also area = (a*b*c)/9 and by using statement A we have area = inradius*(a+b+c)/c.

Using both the equations we can find out value of PR + RQ . Hence option C.

VIEW SOLUTION

37. **A**

Let the purchase price be P and sale price be S. So total brokerage = 0.01P + 0.01S.

If we consider statement A we have S = 1.05*P.

Hence we can find brokerage value in terms of P.

then total profit = S - brokerage = Constant * P.

Then profit per rupee would be dividing last equation by P

Hence statement A is sufficient.

Using statement B we cant make out the profit percentage.

Hence , option A.

VIEW SOLUTION

38. **C**

We have (2 + 0) + (-5 + -6)

Now using statement B it can be written as (0 + 2) + (-5 + -6).

Using trhe statement A we can get the answer to the required question.

Hence,the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

VIEW SOLUTION

39. **D**

Using option B we know that since the constant terms are non-zero and unique , the lines do not intersect at the origin. Also since other coefficients are not equal , the lines can be parallel or intersecting . Hence, the question cannot be answered even by using both statements together.

VIEW SOLUTION

40. **B**

Using statement A we know that C is midpoint of both BD and AE . Also using statement B we know that C is midpoint of BD and AE. Hence , the question can be answered by using either statement alone.

VIEW SOLUTION

41. **D**

Mumbai and No-man's-land would be in different time zones. Using both the statements together we would just know the time mumbai local time when ghosh babu reaches at island. Hence, the question cannot be answered even by using both statements together.

VIEW SOLUTION

42. **D**

According to statement A, |X - Y| = 6

According to statement B, XY = 6k

we have 2 equations and 3 variables, which cannot be solved => Cannot be answered using both the statements.

VIEW SOLUTION

43. **D**
Considering statement B there are 2 possible score values before last 5 mins possible. Korea 3:1 India or Korea 2:0 India. Now considering statement A , in one case India wins (as the score becomes India 4:3 Korea) and in other there is a draw(as the score becomes India 3:3 Korea). Hence, The question cannot be answered by either of the statements.

VIEW SOLUTION

44. **A**
Considering statement A 12 students can be written as 4 students + 8 students. Hence Even if only 4 students are added the teacher can divide students in teams of 8. Now considering statement B , we cant perfectly tell if after adding teacher can divide in teams of 8 . Hence only A is enough to answer the question. Hence , The question can be answered by one of the statement alone but not by the other.

VIEW SOLUTION

45. **A**
Consider statement 1:$(x+y)(1/x+1/y)=4$

$(x+y)^2/xy=4$

$x^2+y^2+2xy=4xy$

$x^2+y^2+2xy-4xy=0$

$(x-y)^2=0$

x = y.

Consider statement 2:$(x-50)^2=(y-50)^2$

(x-y)(x+y-100) = 0

Either x = y or x+y = 100.

Statement 1 is sufficient whereas statement 2 is not sufficient to answer the question.

VIDEO SOLUTION

46. **A**
Let the initial selling price be s1 and changed value be s2, also wholesale cost price be c. According to the given condition we have s1=1.28*c. By statement A we have s2=0.9*s1 and s2-c = 10 . Hence, wholesale cost price can be found out.
Considering statement B we have just s2=50. Using this we can't find c. Hence, the question can be answered by one of the statements alone but not by the other.

VIEW SOLUTION

47. **D**

Consider statement A alone and assume that half of the people scored 600 and half of the people scored 499, so average is not 500

Also there is a possibility that average can be 500 when half of the people score about 500 and the other half score below 500.

Hence we won't get unique answer using A.

Consider statement B alone: We can't find average just by knowing the highest and the lowest score.

Even by using both the statements, we cannot find the answer to the question. Option d) is the correct answer.

VIEW SOLUTION

48. **B**

Statement A: $-1<x<1$
$-3<x-2<-1$
So $1<mod(x-2)<3$
It is definitely greater than 1

Statement B: $-2<x-1<2$
$-3<x-2<1$
which implies mod(x-2) might be greater than or less than 1

So only one statement is sufficient.

VIEW SOLUTION

49. **C**

Let x,y,z be the no. of people speaking only French, only Russian and both respectively. By statement A we have $x+y+z = 300$ and $z = 196$. Using this we can't find number of people in a club who speak only French.
Now considering statement B we have $x = 58$. So using both the statements together we can find number of people in a club who speak only French. Hence, option C is the correct answer.

VIEW SOLUTION

50. **C**

Let amount received by Jagdish, Punit and Girish be j,p,g rs. such that $j+p+g = 38500$. According to statement A , $j= 2*(p+g)/9$. Using this alone we can't answer the question.
Now consider statement B , we have $p = 3*(j+g)/11$. Using this alone we can't answer the question. But considering both statements together we can answer the question. Hence, option C is the correct answer.

VIEW SOLUTION

51. **B**

Given: $P - R = 2370$
or $P = 2370 + R$;
So if we consider only 1st statement, R can be 2 or 6 as 2372 and 2376 both will be divisible by 4.
And if we consider only 2nd statement, R will be having an absolute value of 6 as 2376 will be divisible by 6.

Hence by taking only 2nd statement into consideration, we can find unique value of R.
So answer will be B).

Concept: Try to find a way with which we can find an absolute/unique answer.

VIEW SOLUTION

52. **C**
Concept: Try to see whether we can have a "UNIQUE/ABSOLUTE" answer to given question by applying statements with 3 posibilities.
If only 1st statement is taken into consideration.
If only 2nd statement is taken into consideration.
If both statements are taken into consideration.
In given question with first possibility, we can't have a unique answer as there can be 3 cases. Childs greater than 5 years, childs equal to 5 years and childs less than 5 years and with first possibility we know only that childs greater than 5 years get 5 choclates nothing about other two cases.
With second possibility too, we can't find a unique answer as we can't say anything about childs greater than 5 years
Now with third possibility, we are having a unique answer to given question because we have information about all childs which are either greater than 5 year , lesser or equal to 5 years. Childs older than 5 years will be 5, childs equal to 5 years are 2 and childs younger than 5 years will be 1.

So answer will be C)

VIEW SOLUTION

53. **C**
1. If only 1st statment taken into consideration:
$y = 0.30x$ (Where y= distance from Madras to Trivandrum and x= distance from Calcutta to Madras)
Nothing absolute can be said about average speed from Madras to Trivendrum.
2. If only 2nd statement taken into consideration:
$\frac{y}{t_y} = 2\frac{x}{t_x}$ (Where t_x and t_y are time taken for calcutta to madras and madras to trivendrum respectively.)
And nothing absolute can be said about average speed from Madras to Trivendrum.
3. Now when both statements taken into consideration:
We will have $y = 0.30x$; $\frac{y}{t_y} = 2\frac{x}{t_x}$ and $\frac{x+y}{t_x+t_y}$ = 40 (Given)
After solving above three equations we can find a unique value for average speed from Madras to Trivendrum i.e.
$\frac{y}{t_y} = \frac{920}{13}$
Hence answer will be C.

VIEW SOLUTION

54. **C**
If only first statement taken into consideration:

y-x = 2 where x, y and z are three odd integers.

Nothing absolute can be said about divisibility of x+z with 4.

Now if only second statement taken into consideration :

z-y=2 and nothing absolute can be said about divisibility of x+z with 4.

If we consider both statement simoultaneously then we get x+z = 2y

Now as we know y is an odd integer , so 2y will never get a factor of 4.

Hence x+z is not divisible by 4.

As we find an absolute answer by using both statements Hence answer will be C)

VIEW SOLUTION

55. **D**
Considering first statement only,current unit price of P1 is twice that of P2 and, Rates of P1 and P2 are either decreasing or P1 has constant rate. Now we can't say that which product will be costlier aftre 5 years as nothing certain information is there about rates of decreasing. Similarly if we consider only 2nd statement or consider both statements simultaneously, nothing certain can be said about rate of decreasing hence D will be answer

VIEW SOLUTION

56. **B**
Taking only first statement into consideration:
Nothing absolute can be said about Z and X
Now taking only second statement into consideration:
$W \leq V$
As it is given that X is older than Y; Z is younger than W and V is as old as Y.
and through second statement we know that W is younger or as old as V.
So using above four information we can conclude that Z is not older than X.
Z is either younger or as old as X.
Hence answer will be B).

VIEW SOLUTION

57. **C**
As it is clear that total time of journey will be = Time taken at stop-overs + Time taken while travelling with avg. speed 66 km/h
Hence both informations are needed to solve the question. So our answer will be D.

VIEW SOLUTION

58. **B**
Given equation can be resolved to $\frac{1}{\frac{1}{x}+\frac{1}{y}}$ ($xy \neq 0$ and $x \neq y$)
Now for $\frac{1}{\frac{1}{x}+\frac{1}{y}}$ to be greater than 1,
1/x + 1/y has to be less than 1.
For this to be true both x and y should be greater than 2.
Statement I doesn't give any information about this,
but statement II clearly specifies this. Hence, only
statement II is required to answer the given question.
So it can be answered using statement 2 alone.

VIEW SOLUTION

59. **C**
Taking only 1st statement into consideration, nothing certain can be said about product Q
Similarly with the only 2nd statement, we have only 1 information about product Q and R.
But considering both statements together we can say with sufficient information that product R will be more beneficial than product Q

VIEW SOLUTION

60. **B**
Using first statement alone, we won't be able to find distance as insufficient data is there.
Now using second statement alone, we can find distance d as follows:
velocity before engine failure = $\frac{40}{2}$ = 20 km/h and after engine failure = 5 km/hr
And we know total time exceeds 45 min.
Hence 2 + $\frac{d}{5}$ = $\frac{40}{60}$ + $\frac{40+d}{20}$
or $d = \frac{40}{9}$ km.

VIEW SOLUTION

61. **A**
Considering 1st statement alone, $x^2 + x$ is greater than 50, a two digit number and x is a prime number, then it has to be 7.

On the other hand, we cannot uniquely determine the value of x using statement B alone. Option a) is the correct answer.

VIEW SOLUTION

62. **A**
As Avg. of 3 quotations is 110, so the sum of them will be 330
Now considering 1st statement alone, if the lowest quotation is 100, then highest can not exceed 129 because then the sum of all three will exceed from 330.
So we can get the answer by considering the first statement only.

VIEW SOLUTION

63. **B**
Considering first statement only, given information will not suffice to answer that how many people read both Indian express and Times of India as data regarding intersection of all and data regarding number of people reading other 2 newspapers is not given
Now if we consider only second statement then we can answer about people reading both Indian express and Times of India as follows:
Total readers = 200 = 100 (indian express) + 120 (Times of india) + 50 (Reading none) - Reading both.
So Reading both newspapers = 270-200 = 70

VIEW SOLUTION

64. **A**

Given information X=3(Y-3)
Now considering first statement alone, we will have following equation:
Y+17 = X
On solving two eq. we can find the value of X.
Hence answer will be A.

VIEW SOLUTION

65. **D**
Considering the first statement only, we will get radius > 14. Hence, we can't find the value of radius with this information only.
Now considering the second statement only, we will get radius <= 16.Hence, we can't find the value of radius with this information only.
Now considering both statements simultaneously, we can't find an absolute value of radius.
Hence, answer will be D

VIEW SOLUTION

66. **C**
Considering first and second statement alone, we can't say about time difference as information will not be sufficient.
Now considering both statements together, we can say that if journey duration is of 5 hours and flight departure time is 9:00 a.m., arrival time according to new york will be 2:00 p.m. but it is given that in London, arrival time is 10:00 a.m. Hence, the time difference will be of 4 hours between NewYork and London.

VIEW SOLUTION

67. **C**
Considering first statement and second statement alone, information will not suffice to find the sequence of stations,
But if we consider both statements together we can tell about the sequence of stations as between A and B, Mr. Thomas travels with Mr. Murthy but not with Mr. Rahman. That means Mr. Thomas must have departed from station C and arrived at station D.
And it is also given that between C and D there are two segments. Hence, E lies between C and D, and Mr. Rahman departed from station D.

VIEW SOLUTION

68. **B**
Considering 1st statement alone:
When total number of widgets is 20000
let's say share of type-1 = x
and share of type-2 = y
So x+y = 20000
and 1.1x+0.94y = 20000
Hence we can find value of x and y by statement 1 alone.
Now considering second statement alone;
$x + y = 20000$
$\frac{x}{y} = \frac{2}{1}$

Hence with second statement also, we can find value of x and y
So answer will be B.

VIEW SOLUTION

69. **D**
Considering first statement only, nothing absolute can be said about the value of $a^3 + b^3$ similarly considering 2nd statement only nothing absolute can be said about the value of $a^3 + b^3$.
When we consider both statements together two values of (a+b) can be possible as $\sqrt{28}$ and $-\sqrt{28}$
By which two possible values of $a^3 + b^3$. Nothing absolute can be said about it.
Hence answer will be D).

VIEW SOLUTION

70. **B**
99 = 9 * 11 => If the number is divisible by 9 and 11, then the number is divisible by 99.

=> We can answer the question using only statement I.

The divisibility rule of 9 is that the sum of the digits in the number must be divisible by 9.

The divisibility rule of 11 is that the difference of sums of digits in odd places and digits in even places must be a multiple of 11.

According to statement II, if the number is reversed then the number is divisible by 9 and 11 => Divisible by 99.

Now, on reversing the number again, the number is still divible by 99 because the sum of the digits remains the same and also the difference of sums of digits in odd places and digits in even places remains the same.

Hence, the question can be answered using either of the statements alone.

VIEW SOLUTION

71. **B**
Considering first statement alone, we can calculate 1/3rd of the complete distance as $\sqrt{1^2 + 3^2}$ Hence, we can evaluate complete distance too.

Considering second statement alone, we can calculate 2/3rd of the complete distance as $\sqrt{2^2 + 6^2}$ Hence, we can evaluate complete distance too.

So answer will be B as complete distance can be calculated by using any of the statement alone.

VIEW SOLUTION

72. **D**
Considering first statement alone,
As there are 2 variables x and y, and one equation we can't say anything absolute about values of x and y.
Considering second statement alone,
With 2 variables and one equation, we can't say anything absolute about variables x and y.
Even considering both statements together, we can't find values of x and y as equation in 2nd statement is same as in 1st statement after multiplying it 3.5

VIEW SOLUTION

73. **D**
Considering first statement alone:
Nothing absolute can be said about the hats of Q and R, as Q might have black hat or white hat similarly R might have white or black hat as seen by P.
Considering second statement alone:
Nothing absolute can be said about the hats of P and R as they can have either Black or White caps
Even considering both statements together, we were not be able to reduce the possibilities of having black hats and white hats
Hence answer will be D.

VIEW SOLUTION

74. **C**
Let the speed of the car be x.

Using both the statements, we get a quadratic equation in x.

This equation will have 2 roots, one positive root and one negative root. So, we have to consider only the positive root.

The question can, therefore, be answered by using both the statements together.

VIEW SOLUTION

75. **B**
Considering 1st statement alone:
As height is reduced to 1/3rd of it, radius will also get reduced 1/3rd of complete cone
Now we know volume is directly proportional to product of square of radius and height , we can calculate ratio of volumes.
Similarly considering 2nd statement alone, we can calculate ratio of volumes as we will have sufficient information regarding radius and height.
So answer will be B

VIEW SOLUTION

76. **C**
Considering 1st statement alone:
We can find only one coordinate regarding both lines, which is insufficient information to calculate area.
Considering 2nd statement alone:
We have only one coordinate , regarding one line with insufficient information to calculate area.
Now if we consider both statements together, we can find points where lines are meeting co-ordinate axis and then area.
So our answer will be C.

VIEW SOLUTION

77. **D**

Let the cost price of the Chair be x and that of the table be y.
Using statement 1)
Hence we get the total selling price = 1.15x + 1.2y and profit = 0.15x + 0.2y
Using this alone we can't find the cost price of the chair.
Using Statement 2)
Total cost price = 1.10 x + 1.2y
Using Statement 1 and 2 together,
We get profit in second case = 0.05x.
Reduction in profit = (0.15x + 0.2y) - 0.05x = 0.1x + 0.2y = 20
Hence we can't deduce the value of x even after using both statements together.

VIEW SOLUTION

78. **D**
From Statement 1, we know the speed of each runner.

From Statement 2, we can calculate the length of the track as we know the speed of runner every second.

But the answer cannot be determined as we do not know the direction of each runner.

VIEW SOLUTION

79. **A**
Only statement A: Angles OAC, ACB, CBO are right angles => OACB is a rectangle. OC = AB and OC is radius => AB is equal to radius. Hence, we can find the answer using statement A only.

Only statement B: The point B can change even though the point A s fixed at a distance of 5 units on x-axis from origin. Hence, length of AB changes => We cannot find the answer using statement B only.

VIEW SOLUTION

80. **C**
Let's consider the statements one by one.

Statement 1: n is a multiple of HCF(3,5,7,9) = 315. Hence, n can be 315, 630, 945 .... Hence we cannot say if n is odd or even.

Statement 2: 0<n<400, we cannot say if n is odd or even.

Taking both statements together, n=315. Hence, it is odd.

Thus the question can be answered using both statements together.

VIEW SOLUTION

81. **A**
The definition of the given function is expressed in statement 2, where $a \circledast b = \frac{a+b}{a}$

Hence $2 \circledast 3 = \frac{2+3}{2}$.

Hence the answer can be determined by b alone. A does not give any releant information.

VIEW SOLUTION

82. **D**

Let the total questions be x and the questions solved by Radha and Rani be y and z respectively.

According to the first statement, y+z = 20% of x

Accoding to the second statement, y = 5/3rd of z

There are 2 equations and three variables. Hence the solution cannot be determined.

VIEW SOLUTION

83. **C**
Let price of tea be t, coffee be c and cold drink be d.

Statement 1: c=t+5, we cannot infer value of t from this information

Statement 2: c=d-5 and d=3t, we cannot infer the value of t from this information

Using the statements together, c=t+5 and c=3t-5.

=>t+5=3t-5

=>2t=10

=> t=5

Hence, we can answer the questions using both statements together.

VIEW SOLUTION

84. **D**
We cannot uniquely determine a value for a using either statement alone.

Using both statements together, we get an inconsistent set of equations. Hence, there is no solution using both statements together.

Thus, the question cannot be answered using both statements together.

VIEW SOLUTION

85. **A**
Let the number of girls be g and boys be b. Hence, b+g=150

Statement 1: 50g+25b=4900 => 2g+b=196. Solving the two equations, g=46. Hence, answer can be obtained using statement 1 alone.

Statement 2: 30b+30g=4500 => b+g=150. As both the equations are equal, we cannot find a unique solution.

Thus, the question can be answered using one of the statements alone and not the other.

VIEW SOLUTION

86. **A**
From the statements we need to infer which envelope contains L2.

From statement 1, we know that C2 should open E1.

From statement 2, we cannot tell which envelope contains L2.

Hence, the question can be answered using one of the statements alone and not the other.

VIEW SOLUTION

87. **A**
Let's take the two statements individually:

Statement 1: If the second book is in Rack 3, then the fourth book must be in the only remaining odd numbered rack i.e. rack 1. Hence, we can infer the position of book 4 using the first statement.

Statement 2: If the third book is in Rack 2, then the first book must be in Rack 4. However, we don't know anything about the position of Book 4. Hence, we cannot infer the position of book 4 using the second statement.

Hence, the question can be answered using one of the statements but not the other.

VIEW SOLUTION

88. **A**
Statement 1: a = (X+Y)/2 and $a^2 = XY$. Hence, $a = (X + a^2/X)/2$. Hence, we can re-arrange the equation to represent X in terms of a

Statement 2: There is no relation given between X and a. Hence, we cannot find value of X in terms of a.

Hence, the question can be answered by one statement alone and not the other.

VIEW SOLUTION

89. **C**
We know that r1>r2.

Statement 1: $2\pi(r1 - r2)$ = k. We cannot determine r1 from this information

Statement 2: $\pi(r1^2 - r2^2)$ = m. We cannot determine r1 from this information.

Using both statements together, (r1+r2)/2 = m/k. We now have two linear equations with two variables. Hence, we can determine the value of r1 in terms of m and k.

Hence, the answer can be found using both statements together.

VIEW SOLUTION

90. **A**
In either of the statements, a numerical value is not given.

With the given information, we can only find the radius in terms of N and T.

Hence, the radius cannot be determined using both the statements as well.

VIEW SOLUTION

91. **D**
The proportions of liquid A in vessels 1, 2 and 3 are $\frac{3}{8}$, $\frac{2}{5}$ and $\frac{4}{7}$ respectively.

LCM of 8, 5 and 7 is 280 => Assume each container ha 280 litres.

=> Quantity of liquid A in vessels 1, 2 and 3 are 105, 112 and 160 respectively.

=> Total volume of A = 377 litres

Total volume of B = 840 - 377 = 463 litres

Hence, ratio can be found using both the statements and neither statement alone can be used to find the ratio.

VIEW SOLUTION

92. **D**
For (a-b+c) > (a+b-c)
Considering first statement alone
if b<0, we can't say anything affirmative about inequality as information about c is unavailable.
Considering second statement alone,
if c>0, we can't say anything about b.
Now if we consider both statement together
i.e. b<0 and c>0, then term (-b+c) will be positive and additive to a whether term (b-c) will be negative and subtracting from a.
Hence (a-b+c) > (a+b-c).
And answer will be D.

VIEW SOLUTION

93. **D**
$(\alpha^2+\beta^2)$ can be reduced to $(\alpha+\beta)^2-2(\alpha\beta)$
Now as we can see, we need both values of (α+β) and (αβ) to solve the equation.
Hence answer will be D.

VIEW SOLUTION

94. **A**
Considering first statement alone:
When loss is 25% on cost price, selling price will be $\frac{3}{4}$th of cost price but we can't get the absolute value of cost price.
Considering second statement alone:
As this statement can also be derived from 1st statement, but we need some other informations to get value of cost price
So answer will be A.

VIEW SOLUTION

95. **C**
Cosidering first statement alone:
sp = $\frac{6cp}{5}$ (Where cp is cost price and sp is selling price)
But nothing absolute value can be derived from this information only
Now considering second statement alone:
sp = $\frac{5cp}{4}$ (Where sp is selling price and cp is cost price)
cp is equals to 250 rs.
Now we can derive value of sp from above equation.
Hence answer will be C.

VIEW SOLUTION

96. **A**
Considering given statements, we can't find an absolute number of triangles as lines in a plane can be concurrent also with several other possibilities.
So answer will be A.

VIEW SOLUTION

97. **D**
Considering statement 1 alone, together with all the information we also need information about the time period of compounding interest which is given in second statement.
Hence we can find value of assets using both the statements together.
So answer will be D.

VIEW SOLUTION

98. **A**
Anil's age is a prime number in both 1996 and 1998. Hence, Anil's age in 1996 and 1998 are twin primes (Prime numbers that differ by 2). Anil's age in 1996 and 1998 can be (11,13),(17,19) ,(29,31),.......and so on. Since a unique set cannot be determined, option A is the right answer.

VIEW SOLUTION

99. **D**
Using statement A, we cannot infer the number of students in the class.

Using statement B, we can say that the sum of weights of heaviest and the lighest persons is equal to twice the average weight of the class. But we cannot infer the number of students in the class.

Using both the statements as well, we do not have any information through which we can find the answer.

Hence, option D is the answer.

VIEW SOLUTION

100. **C**
Using statement 1, we come to know that if the inner radius is r, then the outer radius is r+1

Using statement 2, Volume = 20 l = 20000cm^3

$\frac{4}{3}\pi(r+1)^3 - r^3$ = 20000

We can calculate r using this equation.

VIEW SOLUTION

101. **C**
Using only statement A, we know that the shape of th field is rectangle. We cannot infer anything else with only statement A.

Using only statement B, we know that C is one of the vertices, but we do not know how many vertices exist.

Using both A and B, we know that the field is rectangular in shape and the length and breadth are 300 metres and 100 metres.

Hence, we can find the value of the diagonal which is the maximum distance between any two points inside the field.

Hence, option C is the answer.

VIEW SOLUTION

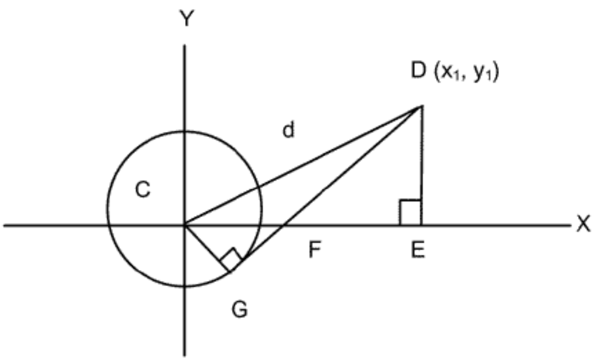
102. **A**
Growth = Change in value on Y-axis / Change in value on X-axis = Tangent of the angle made by the line with the X-axis

In statement 1, since we know both the angles, we can calculate the growth for both the periods and hence the ratio of growths.

Using statement 2 alone, we cannot determine the answer. Option a) is the correct answer.

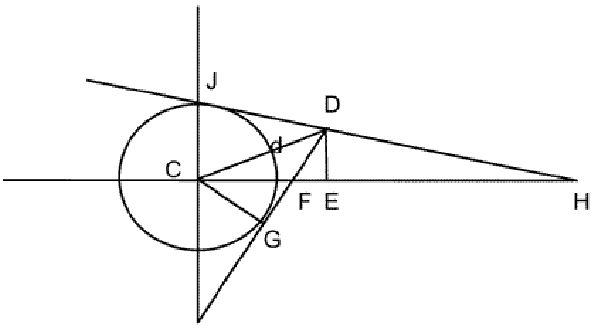
VIEW SOLUTION

103. **B**
From statement I, We can draw the following figure



From the figure we can say that in triangle GCD we know 2 sides GC and CD, so we can find the third side
Also we can find the angle GDC(Using trigonometric ratios)

Similarly in triangle CDE, we know CE and DE, so we can find angle CDE.
From this we can get the angle FDE. Now in right angled triangle FDE, we can find the angle DFE.
So we can answer from statement I
Now let us consider statement II



By using trigonometric relations angle CFG can be calculated.
From this we can get angle DFE
Iin triangle CGD we can find angles GDC(Using trigonometric ratios)
Now GDC = CDJ
Thus we can find EDH and from that we can find EHD.
Hence the question can be answered from statement II alone.

VIEW SOLUTION

104. **A**

ax + by = c

dx + ey = f

Using statement A, we can modify the equation ax + by + c to k(dx+ey=f)

=> We cannot find the point (x,y).

Using statement B, we can change the equations to x + y = c and 2x + 2y = f (f not equal to 2c)

As the slopes are equal and the intercepts are not equal, we can say that the equations do not have a solution.

Hence, we can answer the question using statement B but not A.

VIEW SOLUTION

105. **D**

The value of b is unknown in this question.

Using Statement a, we can say that B is not a mathematician. A may be or may not be a mathematician as we do not know whether he can add three numbers correctly or not. Similarly C may be or may not be a mathematician as we do not know the value of b.

Using statement b, We cannot find out for sure if A and C are mathematicians or not.

So the question cannot be answered even by using both statements together.

VIEW SOLUTION

106. **C**

Using only the first statement, we know that A and B passed the exam. We can't infer anything about C and D.

Using only the second statement, we know that C and D failed the exam. We can't infer anything about A and B.

Using both the statements, we can say that A and B passed the exam and C and D failed the exam.

VIEW SOLUTION

107. **C**

Both x = 4 and x = 16 satisfy the condition in statement A.

Using only statement B, we cannot find the unique value of x.

Using both A and B, we can infer that x = 4

Hence, option C is the answer.

VIEW SOLUTION

108. **D**

Using only statement A, we can say that at least 10 white seeds were used. But we cannot find the exact number.

Using only statement B, we cannot make any inference.

Using both the statements, we can only make the same conclusion as we made using statement A.

Hence, the question cannot be answered by using both the statements together.

VIEW SOLUTION